# Decentralized Coordination of Airport Ground-Support Vehicles with Demand-Weighted Coverage, Buffered Voronoi Safety, and Physics-Consistent Replay

Parham Sharafoleslami
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## Abstract

This paper presents a decentralized coordination framework for autonomous airport ground-support vehicles operating on an apron with time-varying gate service demand. The method combines density-weighted Voronoi coverage control, Buffered Voronoi Cell (BVC) collision avoidance, static gate and parked-aircraft exclusion constraints, and a planar MuJoCo physics backend. Each vehicle computes a local service objective induced by gate demand, projects its desired command through a convex safety filter, and then executes the resulting command through a force/torque dynamics layer. The approach is designed for fuel trucks, baggage tugs, catering lifts, tow tractors, and related ground-service equipment that must distribute across gates without centralized trajectory assignment. We formulate the coverage objective, derive the centroid-seeking control law, state the BVC half-plane constraints, describe the obstacle-aware safety projection, and define the physics-consistent state update used for visualization and analysis. A 16-vehicle, 32-gate replay is evaluated using vehicle separation, gate/aircraft clearance, speed, acceleration, yaw-rate, and heading-consistency metrics. The final trajectory satisfies all vehicle-to-vehicle and gate/aircraft clearance constraints in both sampled frames and linearly interpolated browser playback.

## 1 Introduction

Airport aprons require continuous coordination of heterogeneous ground-support vehicles (GSE), including fuel trucks, baggage tugs, catering lifts, tow tractors, lavatory trucks, and maintenance vehicles. These vehicles must respond to time-varying aircraft service demand while avoiding one another, terminal structures, and parked aircraft. A fully centralized assignment and motion-planning architecture can become brittle as the number of gates and vehicles grows. The project objective is therefore to design a decentralized coordination stack in which vehicles use local geometric information and neighbor interactions to allocate themselves to high-demand gate regions while maintaining collision-free motion.

The framework studied here follows the proposal goals:

- demand-weighted Voronoi coverage for service allocation,
- BVC safety filtering for decentralized collision avoidance,
- time-varying gate demand driven by synthetic or flight-schedule events,
- robust/game-theoretic analysis under demand uncertainty,
- reinforcement-learning extensions for anticipatory positioning, and
- 3D visualization and replay suitable for engineering inspection and presentation.

An important distinction in this implementation is that gate positions are not treated as drivable targets. Gates and parked aircraft are static exclusion regions. Vehicles instead seek apron-side service standoff points associated with gates, and the final MuJoCo-integrated states are projected outside static obstacle regions before export to the web replay.

## 2 Apron and Fleet Model

Let the apron be a compact rectangular domain

$$\Omega = [0, W] \times [0, H] \subset \mathbb{R}^2, \tag{1}$$

where $W = 800\,\mathrm{m}$ and $H = 400\,\mathrm{m}$ in the simplified SFO-like scenario. The fleet contains $N$ ground vehicles with planar positions

$$p_i(t) \in \Omega, \quad i \in \{1, \ldots, N\}. \tag{2}$$

The gate set is

$$\mathcal{G} = \{g_1, \ldots, g_M\}, \quad g_j \in \Omega, \tag{3}$$

with gates arranged on two terminal lines. Since $g_j$ denotes the gate or aircraft service location rather than a drivable parking point, we define a service standoff point

$$s_j = g_j + d_s \sigma_j e_y, \tag{4}$$

where $d_s > 0$ is the apron-side offset, $e_y = [0, 1]^T$, and

$$\sigma_j = \begin{cases} +1, & (g_j)_y < H/2, \\ -1, & (g_j)_y \geq H/2. \end{cases} \tag{5}$$

Thus vehicles are attracted toward service positions near gates, not through parked aircraft or terminal geometry.

# 3 Time-Varying Demand Field

Each gate has a scalar service demand $\phi_j(t) \geq 0$. The simulation uses a base demand plus event-driven exponential decays:

$$\phi_j(t) = \phi_0 + \sum_{\ell:\, k_\ell = j} a_\ell e^{-\lambda_\ell (t - \tau_\ell)} \mathbf{1}_{t \geq \tau_\ell}, \tag{6}$$

where event $\ell$ occurs at gate $k_\ell$, time $\tau_\ell$, magnitude $a_\ell$, and decay rate $\lambda_\ell$. The discrete demand vector is

$$\phi(t) = \begin{bmatrix} \phi_1(t) & \cdots & \phi_M(t) \end{bmatrix}^T. \tag{7}$$

For numerical coverage integration, the domain also includes a low-density background grid $\mathcal{Q}_{\text{grid}}$. The combined discrete coverage domain is

$$\mathcal{Q} = \{s_1, \ldots, s_M\} \cup \mathcal{Q}_{\text{grid}}. \tag{8}$$

# 4 Demand-Weighted Voronoi Coverage

Given vehicle positions $P = \{p_i\}_{i=1}^N$, the Voronoi cell of vehicle $i$ over $\mathcal{Q}$ is

$$V_i(P) = \{q \in \mathcal{Q} : \|q - p_i\| \leq \|q - p_k\|, \ \forall k\}. \tag{9}$$

The coverage objective is the demand-weighted quantization cost

$$H(P, t) = \sum_{i=1}^{N} \sum_{q \in V_i(P)} \frac{1}{2} \|q - p_i\|^2 \rho(q, t), \tag{10}$$

where $\rho(q, t)$ is the demand density assigned to point $q$. For gate service points, $\rho(s_j, t) = \phi_j(t)$; for grid points, $\rho$ is a small background density.

The demand-weighted centroid of $V_i$ is

$$c_i(P, t) = \frac{\sum_{q \in V_i(P)} q\, \rho(q, t)}{\sum_{q \in V_i(P)} \rho(q, t)}. \tag{11}$$

For a first-order point model, the standard Lloyd descent command is

$$u_i^{\text{des}} = k(c_i - p_i), \tag{12}$$

with gain $k > 0$. In the implemented system, this raw command is speed-limited and tapered near the centroid:

$$\tilde{u}_i = \alpha_i u_i^{\text{des}}, \quad \alpha_i = \min \left\{ 1, \frac{v_{\max} \min(1, \|c_i - p_i\| / r_s)}{\|u_i^{\text{des}}\| + \epsilon} \right\}, \tag{13}$$

where $v_{\max}$ is the maximum command speed, $r_s$ is the slow-down radius, and $\epsilon > 0$ prevents division by zero.

# 5 Buffered Voronoi Collision Avoidance

Let $\delta > 0$ be the required vehicle-to-vehicle safety radius. BVC constraints impose that the next point-integrator position of vehicle $i$ remains inside a buffered half-plane relative to every neighbor $j$. For current positions $p_i, p_j$, the BVC half-plane for vehicle $i$ with respect to $j$ is

$$(p_j - p_i)^T p_i^+ \leq \frac{\|p_j\|^2 - \|p_i\|^2 - \delta^2}{2}, \tag{14}$$

where $p_i^+$ denotes the next position of vehicle $i$. Under a velocity command $u_i$ and time step $\Delta t$,

$$p_i^+ = p_i + u_i \Delta t. \tag{15}$$

The safety-filtered command solves the convex quadratic program

$$u_i^\star = \arg\min_{u \in \mathbb{R}^2} \|u - \tilde{u}_i\|^2 \tag{16}$$

$$\text{s.t.} \quad (p_j - p_i)^T (p_i + u \Delta t) \leq \frac{\|p_j\|^2 - \|p_i\|^2 - \delta^2}{2}, \quad \forall j \neq i. \tag{17}$$

## 5.1 Static Gate and Aircraft Exclusion

The initial implementation treated gates only as demand sources, which allowed vehicles to drive through gate markers and parked aircraft in the 3D replay. The corrected model adds circular static exclusions

$$\mathcal{O}_m = \{p \in \Omega : \|p - o_m\| \leq r_m\}, \tag{18}$$

where $o_m$ is a gate or parked-aircraft center and $r_m$ is its exclusion radius. The QP includes a tangent half-plane approximation of the obstacle boundary:

$$n_{im}^T (p_i + u \Delta t - o_m) \geq r_m, \quad n_{im} = \frac{p_i - o_m}{\|p_i - o_m\|}. \tag{19}$$

This keeps the next commanded point on the current safe side of the obstacle.

Because the command filter is applied before physics integration, a final hard projection is applied to the MuJoCo output state:

$$p_i^+ \leftarrow o_m + (r_m + r_b) \frac{p_i^+ - o_m}{\|p_i^+ - o_m\|} \quad \text{if } \|p_i^+ - o_m\| < r_m + r_b, \tag{20}$$

where $r_b$ is a small visualization/safety buffer. Any inward normal velocity component is removed:

$$v_i^+ \leftarrow v_i^+ - \min(0, n_{im}^T v_i^+) n_{im}. \tag{21}$$

## 6 Physics Backend

The high-level planner produces planar velocity targets, but the replayed state is generated through a physics backend. Each vehicle state is

$$x_i = \begin{bmatrix} p_{ix} & p_{iy} & \theta_i & v_{ix} & v_{iy} & \omega_i \end{bmatrix}^T, \tag{22}$$

where $\theta_i$ is heading and $\omega_i$ is yaw rate.

### 6.1 Command Smoothing

Let $u_i^\star$ be the BVC-filtered planar command. The physics layer first applies low-pass smoothing:

$$u_i^{\mathrm{cmd}} = v_i + \beta(u_i^\star - v_i), \quad \beta = \frac{\Delta t}{\tau + \Delta t}, \tag{23}$$

where $\tau$ is the command smoothing time constant.

### 6.2 Force and Torque Control

The desired command heading is

$$\theta_i^{\mathrm{cmd}} = \operatorname{atan2}\left((u_i^{\mathrm{cmd}})_y, (u_i^{\mathrm{cmd}})_x\right), \tag{24}$$

and the wrapped heading error is

$$e_{\theta i} = \operatorname{wrap}(\theta_i^{\mathrm{cmd}} - \theta_i). \tag{25}$$

The target speed is reduced when the desired command direction is far from the current heading:

$$v_i^{\mathrm{tar}} = \min\left\{\left\|u_i^{\mathrm{cmd}}\right\|, v_{\max}\max\left(v_{\min}/v_{\max}, \cos|e_{\theta i}|\right)\right\}. \tag{26}$$

Let

$$f_i = \begin{bmatrix}\cos\theta_i \\ \sin\theta_i\end{bmatrix}, \quad \ell_i = \begin{bmatrix}-\sin\theta_i \\ \cos\theta_i\end{bmatrix} \tag{27}$$

be the forward and lateral directions. The longitudinal acceleration command is

$$a_i = \operatorname{clip}\left(k_v(v_i^{\mathrm{tar}} - f_i^T v_i), -a_{\mathrm{dec}}, a_{\max}\right), \tag{28}$$

and the lateral stabilizing force damps side-slip:

$$F_i = m a_i f_i - m k_\ell(\ell_i^T v_i)\ell_i. \tag{29}$$

The requested yaw rate is clipped by yaw-rate, steering, and lateral-acceleration limits:

$$\omega_i^{\mathrm{tar}} = \operatorname{clip}\left(\frac{e_{\theta i}}{\Delta t}, -\bar{\omega}_i, \bar{\omega}_i\right), \tag{30}$$

where

$$\bar{\omega}_i = \min\left\{\omega_{\max}, \frac{|f_i^T v_i|}{L}\tan\alpha_{\max}, \frac{a_{\mathrm{lat}}}{\max(|f_i^T v_i|, v_{\min})}\right\}. \tag{31}$$

The yaw torque is

$$\tau_i = I_z k_\omega(\omega_i^{\mathrm{tar}} - \omega_i). \tag{32}$$

The vector $(F_{ix}, F_{iy}, \tau_i)$ is applied as generalized force in MuJoCo; the simulator integrates the state and returns $(p_i^+, \theta_i^+, v_i^+, \omega_i^+)$.

## 7 Game-Theoretic and Learning Extensions

The project also includes two higher-level analysis paths.

### 7.1 Worst-Case Demand Game

Let the fleet choose positions $P$ and an adversary choose demand $\phi$ under a budget constraint

$$\phi \in \Phi_B = \left\{\phi \in \mathbb{R}_+^M : \sum_{j=1}^M \phi_j = B\right\}. \tag{33}$$

The robust positioning problem can be written as

$$\min_P \max_{\phi\in\Phi_B} H(P, \phi). \tag{34}$$

The implemented best-response approximation alternates between centroid-seeking updates for the fleet and adversarial demand allocation toward poorly served gates.

### 7.2 MAPPO Extension

For learned anticipatory behavior, each agent receives local observations containing own state, nearby vehicle relative positions, local gate information, and time features. A decentralized actor proposes actions, while a centralized critic uses fleet-level information during training. The BVC filter remains a safety layer applied before dynamics execution.

## 8 Algorithm

One decentralized apron coordination step is:

1. Compute gate demand $\phi(t)$.
2. Build service-domain density on $\mathcal{Q}$.
3. For each vehicle $i$, assign service/grid points to the nearest vehicle, compute the demand-weighted centroid $c_i$, and compute the speed-limited centroid command $\tilde{u}_i$.
4. For each vehicle $i$, solve the BVC and obstacle-aware QP for $u_i^\star$.
5. Apply smoothed force/torque controls in MuJoCo.
6. Project final states outside static exclusion zones.
7. Log positions, headings, speed, yaw rate, demand, and diagnostics.

## 9 Results

The current presentation replay uses $N = 16$ vehicles, $M = 32$ gates, 300 frames, and $\Delta t = 0.5\,\mathrm{s}$. Nineteen gates are populated with parked aircraft in the web replay. The obstacle radii are 16 m for gates and 20 m for parked-aircraft gates, with an additional projection buffer.

Table 1: Final Web Replay Diagnostics

| Metric | Value | Limit/Target |
|---|---|---|
| Vehicles / gates / frames | 16/32/300 | – |
| Minimum vehicle separation | 31.605 m | $> 10$ m |
| Vehicle safety violations | 0 | 0 |
| Minimum gate/aircraft endpoint clearance | 0.750 m | $> 0$ |
| Minimum interpolated segment clearance | 0.733 m | $> 0$ |
| Obstacle penetrations | 0 | 0 |
| Mean speed | 0.834 m/s | – |
| 95th percentile speed | 3.184 m/s | $< 5$ m/s |
| Maximum speed | 4.997 m/s | 5 m/s |
| 95th percentile acceleration | 0.242 m/s$^2$ | $< 1.5$ m/s$^2$ |
| Maximum acceleration | 1.504 m/s$^2$ | 1.5 m/s$^2$ |
| Maximum deceleration magnitude | 3.612 m/s$^2$ | nominal 2.5 m/s$^2$ |
| 95th percentile yaw rate | 0.387 rad/s | $< 0.8$ rad/s |
| Maximum yaw rate | 0.619 rad/s | 0.8 rad/s |

### 9.1 Are the Agent Behaviors Normal?

The final replay is normal with respect to the main proposal goals: vehicles distribute to service demand, avoid one another, avoid gate/aircraft exclusion zones, and produce physically smooth headings under MuJoCo integration. The safety constraints are conservative rather than tight: the minimum vehicle separation is much larger than the 10 m safety radius. This means the scenario demonstrates safe coordination but does not stress the BVC boundary.

Some limits are intentionally reached. The speed limit is reached by a small number of samples, and acceleration reaches the configured upper bound during initial acceleration. Yaw-rate does not hit its configured limit. The one concerning metric is maximum deceleration magnitude, which can exceed the nominal deceleration bound because the post-integration obstacle projection removes inward velocity components. This is a deliberate hard-safety correction for visualization and obstacle non-penetration, but it should be reported as a projection artifact rather than a pure actuator-limited braking event.

## 10 Discussion and Limitations

The current implementation has three important limitations.

First, BVC guarantees are exact for the single-integrator model used in the QP, while the executed dynamics are MuJoCo-integrated. The final state projection ensures visual and geometric non-penetration, but a formal proof for the coupled BVC–MuJoCo system would require a dynamics-aware reachable-set safety filter.

Second, the MuJoCo model is planar and force/torque controlled. It is more physical than direct position interpolation or kinematic replay, but it is not yet a full wheel-contact tire model with steering joints, rolling constraints, tire friction, and actuator dynamics. Such a model is the next step for high-fidelity vehicle behavior.

Third, the present 16-vehicle scenario is safe but not safety-critical: vehicles rarely approach the BVC boundary. A complete evaluation should include a dense stress test in which vehicles must pass through constrained corridors or serve adjacent gates under high simultaneous demand.

## 11 Conclusion

This paper formulated a decentralized airport GSE coordination framework combining demand-weighted Voronoi coverage, BVC collision avoidance, gate/aircraft obstacle exclusion, and MuJoCo-integrated vehicle dynamics. The corrected simulation no longer drives vehicles through gate or aircraft markers; service targets are offset from gates, static obstacles are included in the safety filter, and final physics states are projected outside exclusion zones before visualization. The latest 16-vehicle, 32-gate replay satisfies vehicle separation and obstacle-clearance requirements in both sampled states and browser-interpolated playback. The resulting behavior is consistent with the project proposal, while highlighting future work on dynamics-aware safety guarantees and full wheel-contact MuJoCo vehicle models.

## References


[1] J. Cortes, S. Martinez, T. Karatas, and F. Bullo, "Coverage control for mobile sensing networks," *IEEE Transactions on Robotics and Automation*, vol. 20, no. 2, pp. 243–255, 2004.

[2] D. Zhou, Z. Wang, S. Bandyopadhyay, and M. Schwager, "Fast, on-line collision avoidance for dynamic vehicles using buffered Voronoi cells," *IEEE Robotics and Automation Letters*, vol. 2, no. 2, pp. 1047–1054, 2017.

[3] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.

[4] C. Yu et al., "The surprising effectiveness of PPO in cooperative multi-agent games," in *Advances in Neural Information Processing Systems*, 2022.